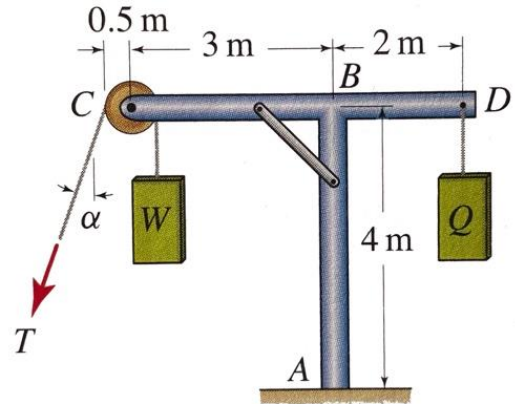


Exam 3: Practice Exam

1. (4 pt) Determine the magnitude of the shear force at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 98.4 N
- B. 92.1 N
- C. 87.6 N
- D. 82.5 N
- E. 79.3 N



2. (4 pt) Determine the magnitude of the normal force at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

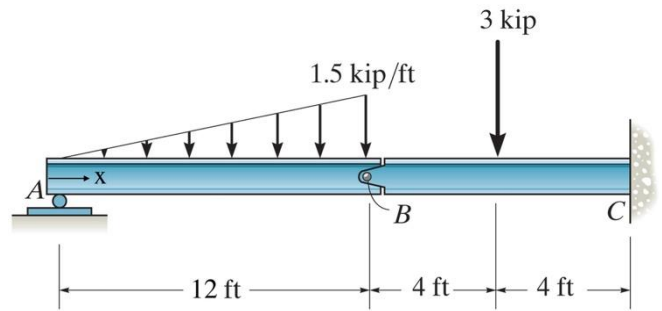
- A. 356 N
- B. 362 N
- C. 368 N
- D. 374 N
- E. 383 N

3. (5 pt) Determine the magnitude of the bending moment at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 829 N-m
- B. 838 N-m
- C. 846 N-m
- D. 812 N-m
- E. 801 N-m

Section 2 – Shear and Moment Diagrams

Provide information for constructing the **shear diagram**. The x origin is on the left at A and the support reactions are $A_y = 6$ kip acting up, $C_y = 15$ kip acting up and $M_C = 108$ kip-ft acting clockwise. A is a roller, B is a pin and C is a fixed support.



4. (3 pt) What happens to the shear diagram at $x = 12$ ft?
- | | |
|-------------------|---------------------|
| A. No change | E. Decrease 3 kip |
| B. Decrease 9 kip | F. Increase 3 kip |
| C. Increase 9 kip | G. Decrease 1.5 kip |

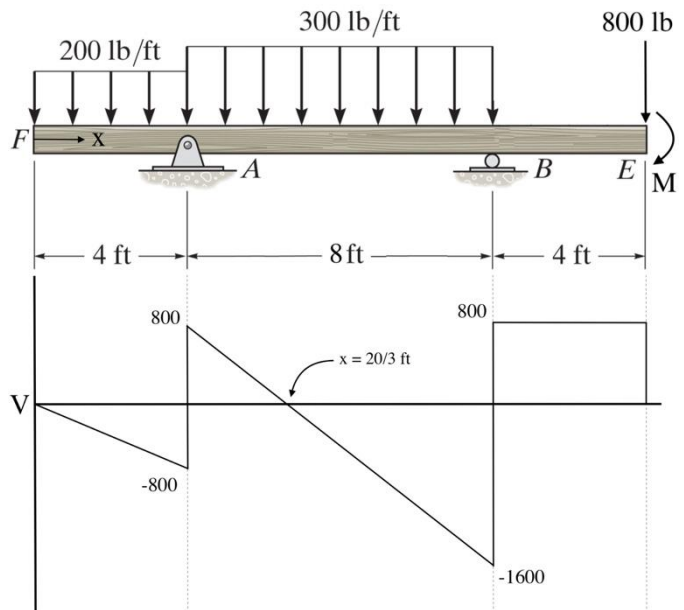
Determine the numerical value (magnitude only) of the change in shear for section $0 < x < 12$ ft as well as the behavior (shape) of the shear for the same region.

5. (3 pt) Change = _____ kip (magnitude only)
- | | |
|--------|-------|
| A. 9 | D. 12 |
| B. 1.5 | E. 18 |
| C. 6 | F. 0 |

6. (3 pt) Behavior: A B C D E F G

7. (3 pt) What happens to the shear diagram at $x = 16$ ft?
- | | |
|-------------------|-------------------|
| A. Decrease 3 kip | D. Increase 9 kip |
| B. Increase 3 kip | E. Decrease 6 kip |
| C. Decrease 9 kip | F. Increase 6 kip |

Provide information for constructing the **moment diagram**. The x origin is on the left at A and the support reactions are $A_y = 1600$ lb acting up, $B_y = 2400$ lb acting up. A is a pin and B is a roller. The couple $M = 1600$ lb-ft CW at E.



Determine the numerical value (magnitude only) of the change in moment for section $4 < x < 6.667$ ft as well as the behavior (shape) for the same region.

8. (3 pt) Change = _____ lb-ft (magnitude only)
- | | |
|---------|---------|
| A. 1067 | D. 2400 |
| B. 800 | E. 4267 |
| C. 1600 | F. 3200 |

9. (3 pt) Behavior: A B C D E F G

10. (3 pt) What happens to the moment diagram at $x = 16$ ft?

- | | |
|------------------------|------------------------|
| A. Increase 1600 lb·ft | D. Decrease 1600 lb·ft |
| B. Decrease 400 lb·ft | E. Increase 800 lb·ft |
| C. Increase 400 lb·ft | F. No change |

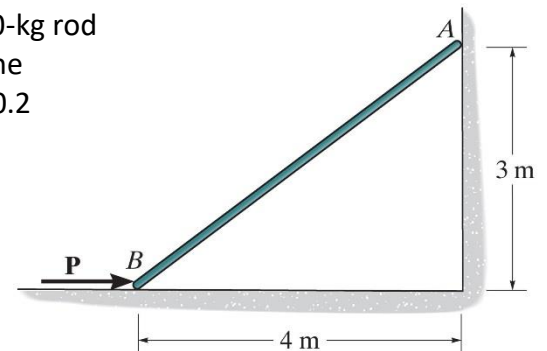
11. (4 pt) Determine the order of the equation for the internal bending moment for the region $12 < x < 16$ ft on the beam shown.

- A. third order, cubic
- B. constant
- C. second order, parabolic
- D. zero
- E. first order, linear

Section 3 - Friction

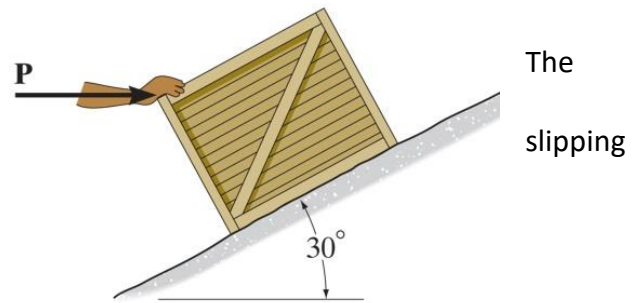
12. (5 pt) Determine the minimum force **P** to prevent the uniform 30-kg rod *AB* from sliding down. The contact surface at *B* is smooth, whereas the coefficient of static friction between the rod and the wall at *A* is $\mu_s = 0.2$

- A. 143 N
- B. 155 N
- C. 138 N
- D. 129 N
- E. 117 N



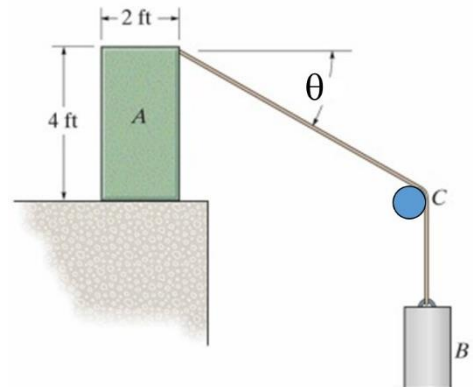
13. (5 pts) Determine the minimum horizontal force **P** required to prevent the crate from slipping **down** the plane. The crate has a mass of 50 kg and the coefficient of static friction between the crate and the plane is $\mu_s = 0.25$. At the point of slipping downward, the normal force at the contact surface is 495 N.

- A. 140 N
- B. 203 N
- C. 449 N
- D. 474 N
- E. 335 N



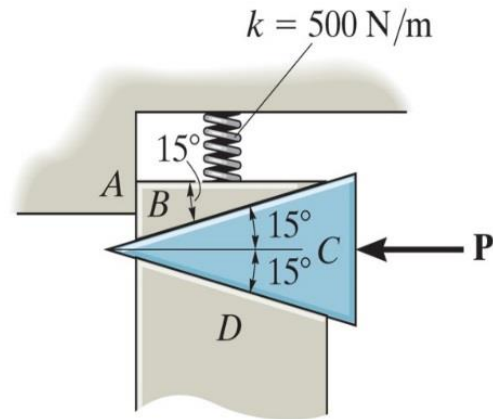
14. (5 pts) If $\theta = 60^\circ$, what is the value of β when using the belt friction equation for the cord at peg C?

- A. 1.05 radians
- B. 0.52 radians
- C. 0.79 radians
- D. 1.12 radians
- E. 0.68 radians



15. Force P is applied to cause motion. On a free-body diagram of wedge B , the direction of the friction force at the contact surface between B and C would be drawn....

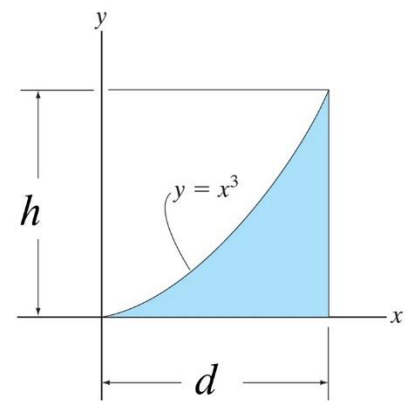
- A. down and to the right
- B. up and to the left
- C. down and to the left
- D. up and to the right



Section 4 – Centroids A

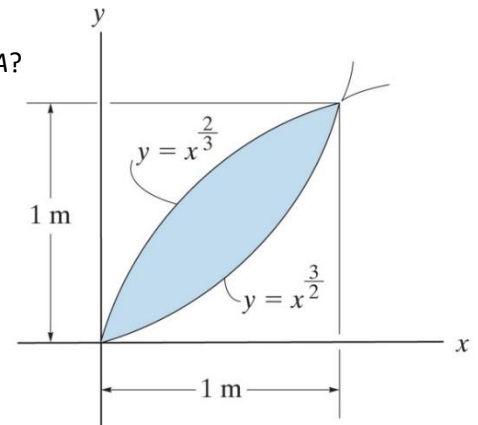
16. (5 pt) Determine the location of $\bar{x}$ of the shaded area. $d = 1.5$ m
The area of the shaded part is 1.266 m^2

- A. 1.04 m
- B. 0.96 m
- C. 1.20 m
- D. 0.83 m



17. (5 pt) When using a **vertical** differential element of area to solve for the location of $\bar{x}$ for the shaded area what expression would be correct for dA ?

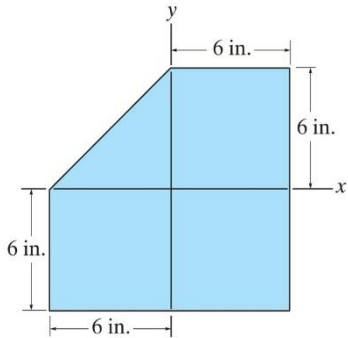
- A. $(x^{6/7} + x^{1.5})/2 \, dx$
- B. $(x^{6/7} - x^{1.5}) \, dx$
- C. $(x^{6/7} + x^{1.5}) \, dx$
- D. $(x^{6/7}) \, dx$



18. (5 pt) Determine the location of $\bar{y}$ for shaded area. *Note location of the origin.*

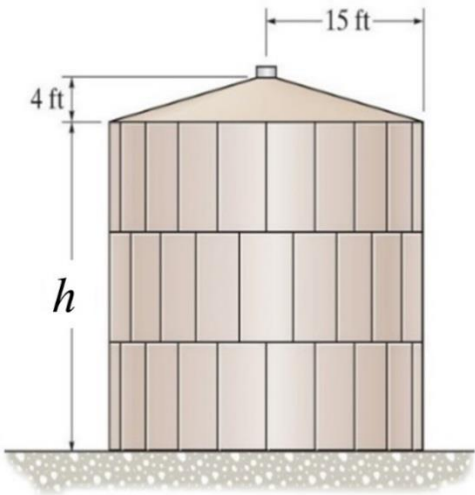
- A. -0.57 ft B. -0.63 ft C. -0.75 ft D. -0.86 ft

Segment			



19. (5 pt) (4 pt) Determine the **volume** of the storage tank. $h = 30$ ft.

- A. 18,614 ft³
 B. 21,012 ft³
 C. 19,803 ft³
 D. 22,148 ft³



20. (5 pt) Determine the **outside** surface area of the storage tank including the roof. $h = 30$ ft. *(do **not** include the floor).*

- A. 3559 ft²
 B. 3253 ft²
 C. 3197 ft²
 D. 3088 ft²